



National University of Modern Languages

Unity Task 1

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Course:	Computer Graphics
Class:	BSCS VII-A
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1. Objective

The objective of this project was to create a simple 3D scene in Unity using basic shapes and materials. The goal was to become familiar with the Unity environment, learn how to structure a scene, apply materials, add lighting, and configure the camera.

2. Tools and Environment

Software Used: Unity Editor version 6.1.0 (6000.1.5f1)

Target Platform: Windows, Mac, Linux

Development Mode: 3D Project Template

3. Steps Followed

Step 1: Project Setup

A new 3D project was created in Unity Hub and named **Basic3DScene**.

The project was saved in the chosen directory and opened for editing.

Step 2: Scene Configuration

A new empty GameObject named **Scene** was added and its transform was reset to (0, 0, 0).

This GameObject serves as the parent container for the entire scene structure.

Step 3: Ground Plane Addition

A **Plane** was added as a child of the Scene GameObject and renamed to **Ground**.

It was scaled to (10, 1, 10) to act as the floor of the scene.

Step 4: 3D Object Creation

A **Cube** was created and added as a child of the Scene GameObject.

The cube was renamed **Cube1** and positioned above the ground to make it visible.

Step 5: Material Application

A **Materials** folder was created in the Assets directory.

A new **Material** named *Green* was created and its Albedo color was set to green.

This material was applied to **Cube1** to change its appearance.

Step 6: Lighting Setup

A **Directional Light** was added to illuminate the scene.

The light was positioned and rotated to cast visible shadows and highlight the 3D object.

Step 7: Camera Adjustment

The **Main Camera** was repositioned and rotated to get a clear perspective of the cube and ground.

The camera's field of view was adjusted for optimal framing.

Step 8: Scene Testing

The **Play** button was used to enter Play mode.

Basic testing was conducted to confirm visibility, lighting, and material appearance.

Step 9: Scene Saving

The scene was saved as **Basic3DScene.unity** within the *Scenes* folder.

Step 10: Build Settings

The current open scene was added to Build Settings.

A build was initiated for the selected platform (e.g., Windows).

4. Observations

All objects rendered correctly with proper material and lighting.

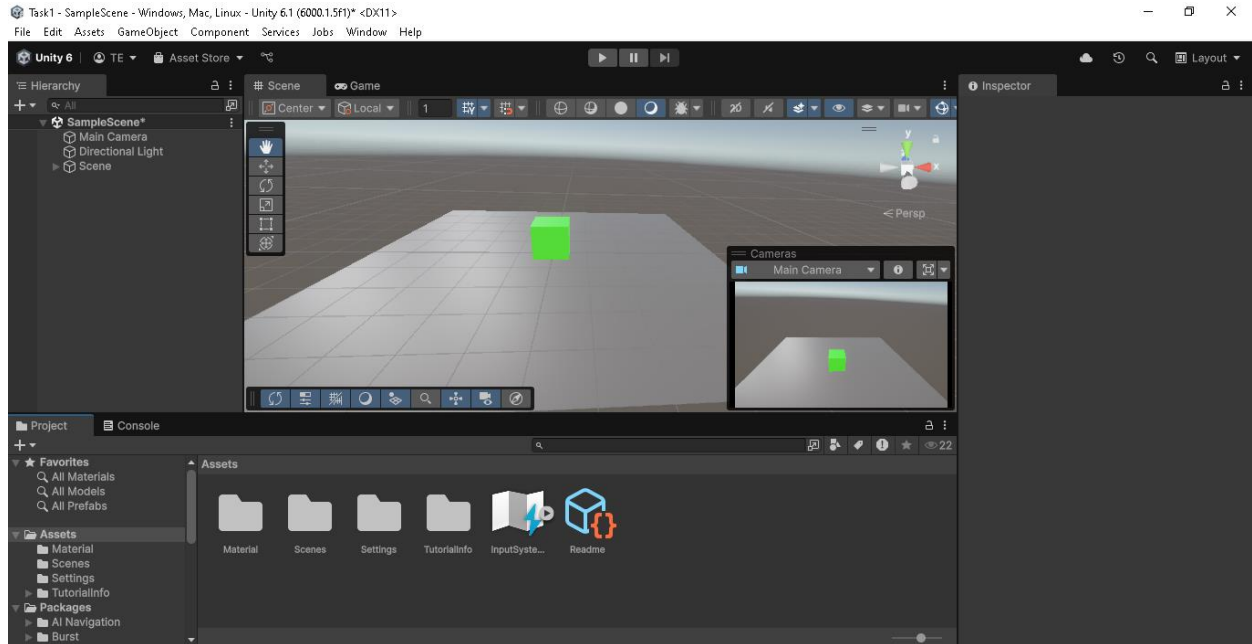
A minor Unity Console message was observed:

Assertion failed on expression: 'f->previewArtifactID == found->second.previewArtifactID'

This is a known benign Unity Editor warning and did not affect runtime performance.

5. Screenshot

Below is a screenshot showing the cube with the green material, the camera preview, and project structure:



6. Conclusion

The objective of creating a simple 3D scene using Unity was successfully achieved. This project provided hands-on experience with Unity's editor, object hierarchy, materials, lighting, and camera tools. The scene is ready for extension and further experimentation with more advanced 3D elements, interactivity, and physics.
